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הפקולטה למדעי המחשב והמידע ע"ש שטיין
אוניברסיטת בן-גוריון בנגב
התוכנית להנדסת תוכנה

# SwipeLab


## Sagi Evroni, Eden Bar, Ofri Hanochi, Avihoo Amos

**Seminar Advisor: Prof. Robert Moskovich | Professional Advisor: Prof. Chen Kaesar**


## BACKGROUND & PROBLEM

**1. Critical Agricultural Threat**

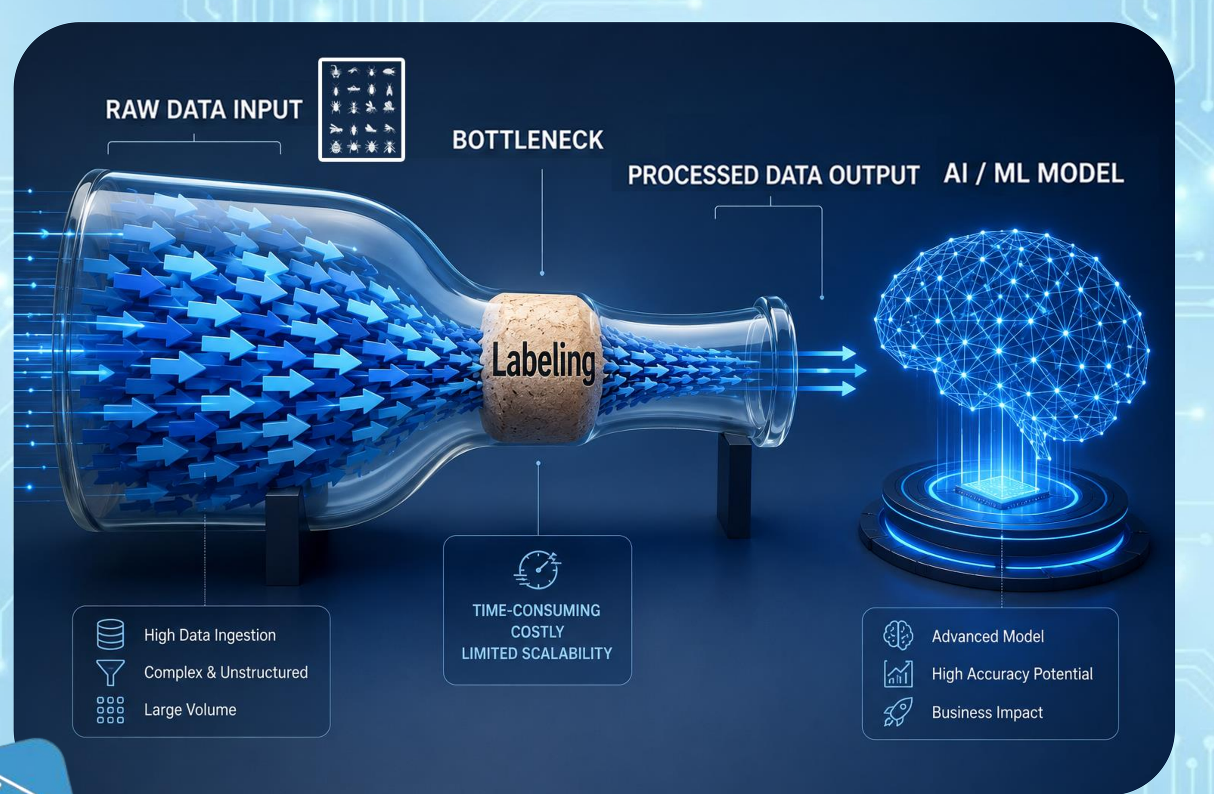
Invasive pests require constant monitoring:

**Trap → Scan → Analyze**

**2. Data Overload**

Traps generate **100,000s of images** that need to be labeled.


**3. The Bottleneck**




## MOTIVATION

**Scale:** To bridge the gap between 380,000+ raw images and a labeled dataset ready for Machine Learning.

**Time-Efficiency:** To reduce reliance on expensive expert hours for routine identification.

**Accessibility:** Harnessing the "wisdom of the crowd" by transforming a tedious scientific task into a mobile-friendly activity for volunteers.

## THE SOLUTION

**Gamified Labeling:** Turning "work" into a mobile "swipe" game available from anywhere at any time.
**Crowdsourced Intelligence:** Scaling expert tasks to the public.
**Rapid Throughput:** Seconds per label vs. minutes.

## RESULTS

**Speed Increase** from 240s/image to 2–4s/image

**App In Use** by 30+ users

**High-Throughput Labeling** with a small volunteer group

## CONCLUSIONS & INSIGHTS

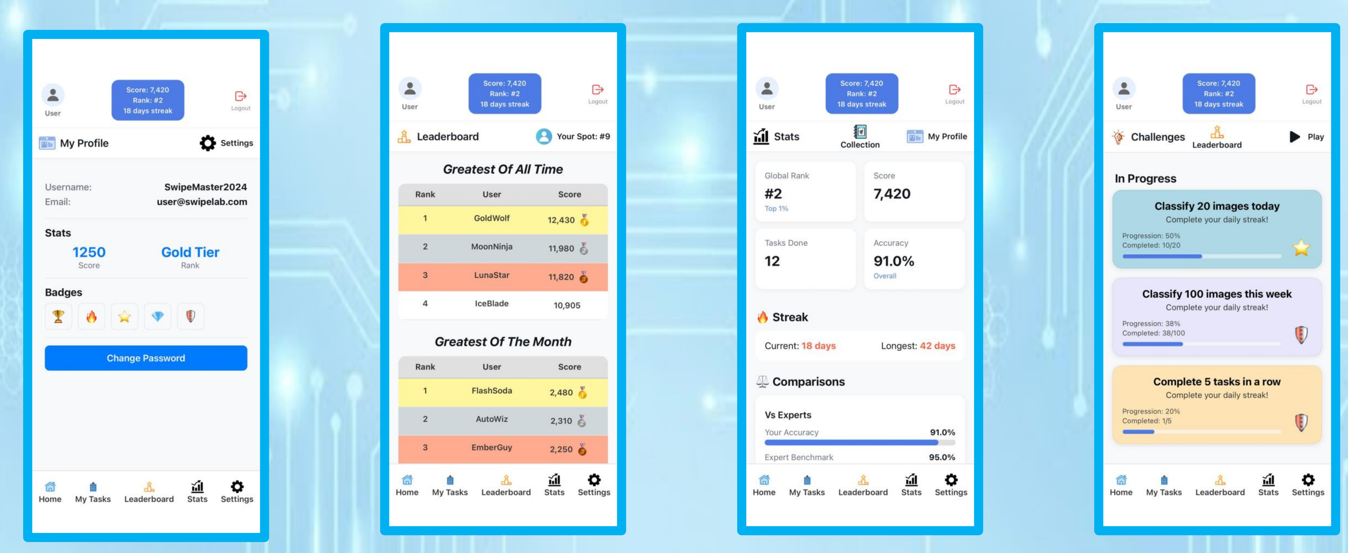
**Gamification Works:** Scientific data labeling can be a competitive game, significantly increasing the volume of processed data.
**Reliability:** Non-experts can provide expert-level data when a robust consensus algorithm is applied.
**Future Impact:** SwipeLab provides the "fuel" for future Deep Learning models that will eventually automate pest detection entirely, saving the agriculture industry millions in potential losses.

## KEY COMPONENTS

**Mobile Interface:** An app designed for "micro-tasking"

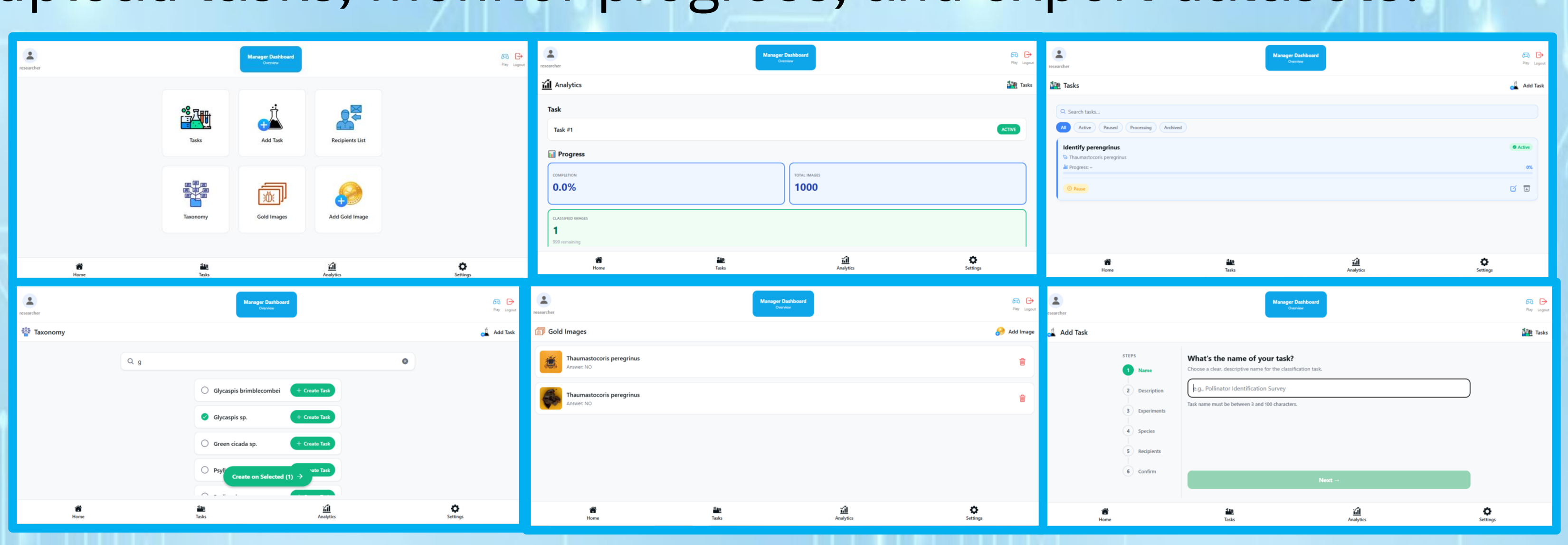

**Gamification Layer:**

The system maintains user engagement through scores, badges, daily streaks, challenges, and competitive leaderboards.



**Researcher Dashboard:**

A web-based management tool where scientists can upload tasks, monitor progress, and export datasets.



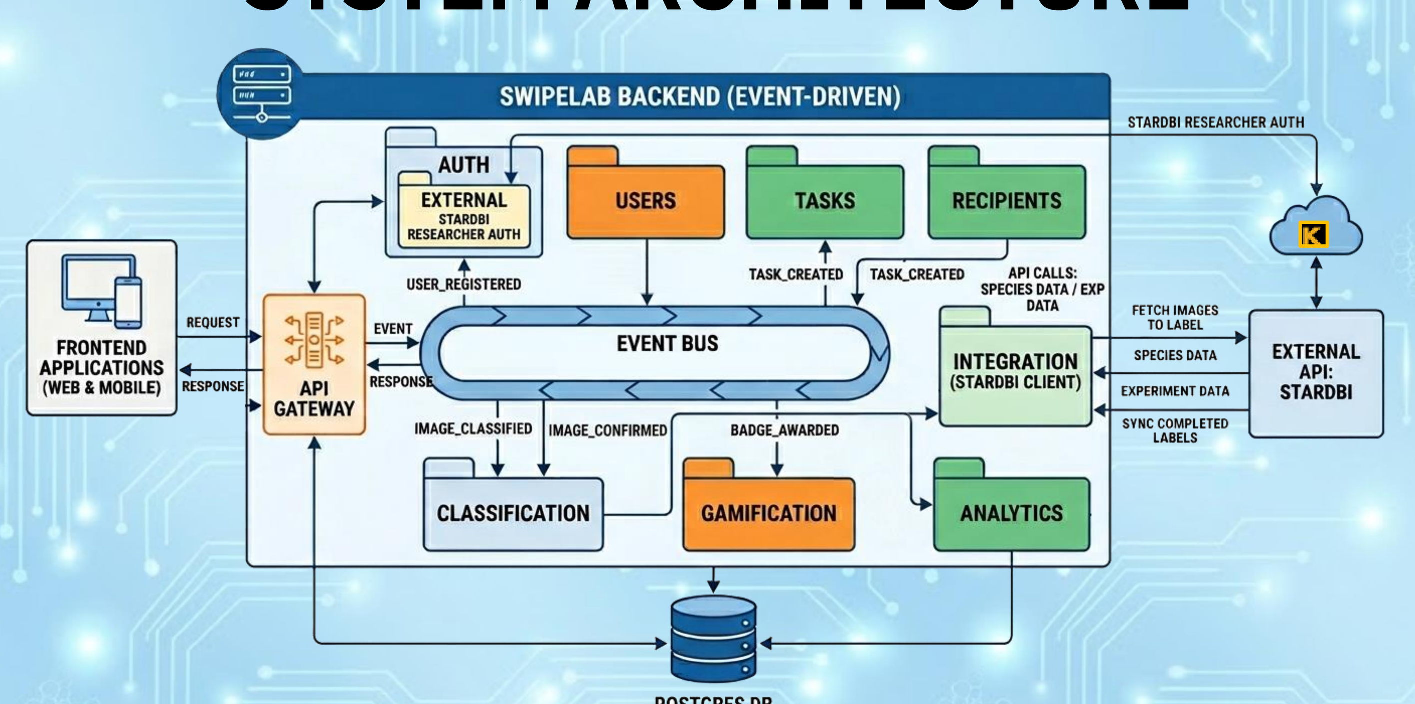
**Improving Trust in Labeled Data:**

**User Credibility:** The system measures user reliability using **Cohen's Kappa** agreement with expert annotations and validation on **"gold standard"** images.

**Crowd Consensus:** User labels are compared against the majority consensus of the crowd to measure consistency and improve overall annotation quality.

**Bot Detection:** Algorithmic filters to prevent "noise" or random swiping.

## SYSTEM ARCHITECTURE




## TECHNOLOGIES

React Native | Expo | spring boot | PostgreSQL | GitHub Actions | docker